

The anchor vectors:

- $\mu_1 = (1.416, 3.304, 2.312) \rightarrow \text{sadness}$
- $\mu_2 = (1.192, 4.560, 2.088) \rightarrow \text{depressed}$
- $\mu_3 = (1.392, 4.376, 2.304) \rightarrow \text{sorrowful}$

These were selected because:

- They represent **core depressive emotions** in the **Valence–Arousal–Dominance (VAD)** space.
- According to **PAD/VAD theory**, depression is typically associated with:
 - **Low valence** (unpleasantness)
 - **Moderately low arousal** (fatigue, lethargy)
 - **Low dominance** (helplessness)
- These values were sourced from the **NRC-VAD Lexicon**, a trusted affective resource.
- Original scores (0–1 scale) were **rescaled to 1–9** using:

$$\phi(u) = 1 + 8u$$

- The selected anchors form a **tight cluster** in VAD space, making them ideal reference points for measuring depressive proximity.

Step-by-Step: How to Calculate Depression Score from VAD

Let's say you have a VAD score:

$$x = (v, a, d)$$

◆ Step 1: Compute Euclidean distance to each anchor

For each anchor $\mu_k = (v_k, a_k, d_k)$, compute:

$$\|x - \mu_k\|_2 = \sqrt{(v - v_k)^2 + (a - a_k)^2 + (d - d_k)^2}$$

Do this for all three anchors.

Step 2: Average the distances

$$D_{\text{multi}}(x) = \frac{1}{3} \sum_{k=1}^3 \|x - \mu_k\|_2$$

Where:

- $d_{\min} = 0.7892$
- $d_{\max} = 9.7557$

◆ **Step 4: Invert the score**

Since **closer distance = more depressive**, invert the normalized score:

$$\text{DepressionScore}(x) = 100 - s$$

✓ **Final Formula Summary**

Given $x = (v, a, d)$:

1. Compute:

$$D_{\text{multi}} = \frac{1}{3} \sum_{k=1}^3 \sqrt{(v - v_k)^2 + (a - a_k)^2 + (d - d_k)^2}$$

2. Normalize:

$$s = \frac{D_{\text{multi}} - 0.7892}{9.7557 - 0.7892} \times 100$$

3. Final score:

$$\text{DepressionScore} = 100 - s$$